

KICC Global Exhibition Platform

Achievement Report — August 2026

1. PLATFORM OVERVIEW

The KICC Global Exhibition Platform is a complete digital ecosystem for Kenya's premier convention venue and all 47 counties. It connects county governments, trade boards, private exhibitors, international buyers, and event organisers on a single unified system. The platform operates as a production-grade Laravel application served through a Cloudflare edge network, with media assets delivered via a global CDN. A Kotlin engine powers the 3D, AI, and marketplace backend.

2. INFRASTRUCTURE & DEPLOYMENT

The platform runs on a production server with PHP 8.4, connected to a TiDB Cloud database for all application data. A Cloudflare Workers edge gateway handles routing, caching, rate limiting, and JWT validation. Media files are stored in Cloudflare R2 object storage and served through a dedicated worker with CORS headers, range request support, and immutable cache rules. The entire system is deployed via Laravel Cloud with automated CI/CD through GitHub Actions. Upload limits have been configured to 1GB to support large video files, and PHP-FPM has been tuned for production performance.

3. 47-COUNTY SYSTEM

All 47 Kenyan counties are fully onboarded with individual public websites, each with sector-specific content, images, product listings, tourism attractions, hotels, and trade data. Counties are organised by region (Central, Coast, Eastern, Nairobi, North Eastern, Nyanza, Rift Valley, Western) with a searchable, filterable directory. Each county has a professional admin dashboard where officials can manage content, upload images, set prices, advertise, manage 4D immersive videos, toggle sectors, add entities, and download reports. The county exhibitor portal provides a separate view for trade board members to track products, exhibitors, orders, and escrow revenue.

4. KICC MOTHER ADMIN

The KICC Mother Admin is a central control panel that oversees the entire platform. From here, administrators can manage: all 47 counties and their regional groupings, national government exhibitors, private exhibitors, marketplace orders, provider certifications, escrow transactions, users, and the landing page hero media. The admin includes an Artisan command console for system maintenance, a provider certification queue, and release controls for escrow funds.

5. MEDIA PIPELINE

Every image, video, and 3D model on the platform is managed through a central MediaAsset system with automatic derivative generation. Files are uploaded to R2 storage and served via CDN. The system supports: video posters, thumbnails, multiple quality variants, WebP conversion, HLS streaming, and 4D immersive video slots. No file paths are hardcoded — everything is resolved through the database at runtime, enabling hot-swappable content through the admin panel.

6. COMMERCE & MARKETPLACE

The platform includes a full marketplace where counties and private exhibitors list products for sale. Each product has variants, pricing, stock tracking, and category associations. Orders are processed through an escrow system that protects both buyers and sellers. The marketplace supports search, filtering, booking flows, and a complete order management system. Product bookings include a confirmation and success workflow.

7. TRADE SYSTEM

A comprehensive trade module connects Kenyan counties to international markets. The system tracks trade agreements across trading blocs including EAC, COMESA, AfCFTA, and WTO. An export eligibility checker

helps sellers determine tariff preferences, and an export enquiry flow creates trade lead records that integrate with n8n workflow automation for follow-up.

8. TRAVEL & TOURISM

Each county features tourism attractions, hotels, cultural sites, and transport information. The system supports tour guide onboarding, car rental listings, restaurant discovery, event organisers, and agent/operator KYC verification. Reviews and ratings with photo uploads are available for all listings, with a moderation queue for administrators.

9. VIDEO & CINEMATIC CONTENT

The landing page hero video showcases the KICC venue through an 81-second cinematic walk-through generated from building photographs. The video features: deep push-in toward the Main Gate (aimed at the gate on the right side), reception-to-concourse walk, grand staircase ascent, Tsavo Hall, Amphitheater montage, Lenana Room, and a brand finale. All scenes include TV-news-style overlays (LIVE badge, lower-thirds, scrolling tickers), Ken Burns slow-zoom effects, and professional color grading. The video is compressed to 21MB for fast streaming and served with CORS headers for cross-origin playback. County admin users can upload their own 4D immersive videos per sector entity through the admin panel.

10. 3D & IMMERSIVE TECHNOLOGY

The platform includes a 3D content pipeline that generates holographic wiggle videos from photo bursts, GLB 3D models, and parallax depth effects. The FX3D JavaScript engine provides tilt parallax, scroll-reveal animations, Ken Burns cinematic zoom, and light-sweep effects on all media cards. A three.js particle-terrain layer enhances the hero section. 3D models are generated from real photo bursts, not synthetic renders.

11. SECURITY & AUTHENTICATION

The platform implements email-based verification codes for registration and login, Google OAuth for social login, MFA/TOTP via Google Authenticator, role-based access control (kicc_admin, county_admin, exhibitor, provider), Sanctum API tokens for mobile apps, and Passport OAuth2 for third-party integrations. A rate-limiting edge gateway protects auth endpoints, and all sessions are secured with proper CSRF protection.

12. AI & INTELLIGENCE FEATURES

The platform integrates AI capabilities including: image search via OpenRouter vision models (Qwen-VL), a chatbot for visitor assistance, personalised recommendations, tourism trend analytics, demand forecasting, fraud detection, and embedding-based content search. An analytics dashboard tracks tourism intelligence metrics.

13. API & INTEGRATIONS

A public OpenAPI specification is exposed at /api/openapi. The platform integrates with: SendGrid for transactional email, n8n for workflow automation (provider approvals, trade enquiries), weather data for each county, GIS mapping, USSD for mobile feature phones, Google OAuth, and Cloudflare services for edge caching and media delivery. Webhooks notify external systems of platform events.

14. TESTING & QUALITY

The codebase includes 148 automated tests with 202 assertions covering API endpoints, authentication, CORS, USSD, search, and security. A penetration testing suite validates backend security. Database scalability indexes have been added for million-row tables. Laravel Horizon manages production queues, Pulse monitors performance, and Sentry tracks errors in production.

15. KEY ACHIEVEMENTS SUMMARY

- Platform: Live production Laravel application serving kicctest.org through Cloudflare edge network.

- Counties

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